

Distance Control Light

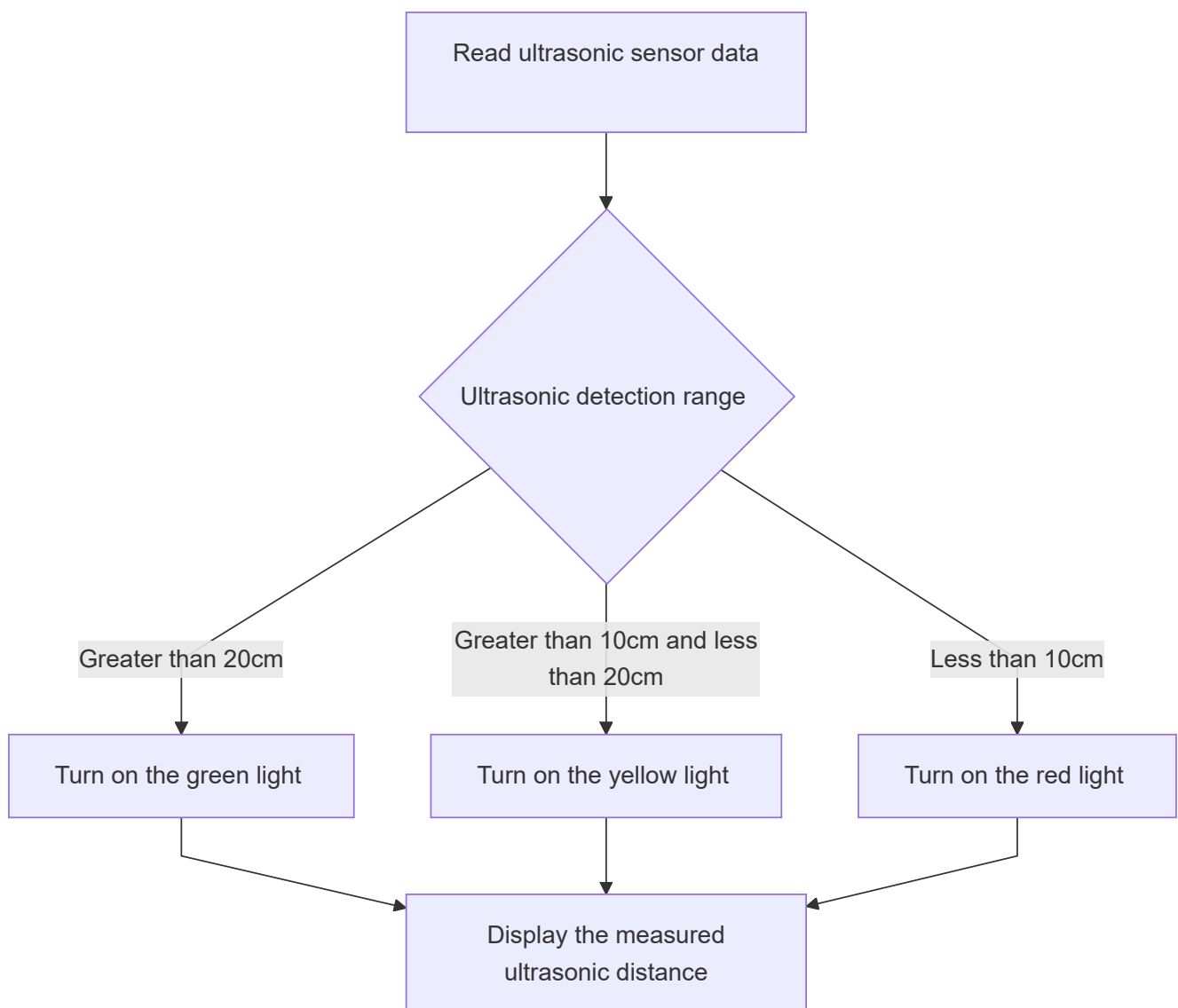
Distance Control Light

1. Experiment Purpose
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5. Experiment Phenomenon

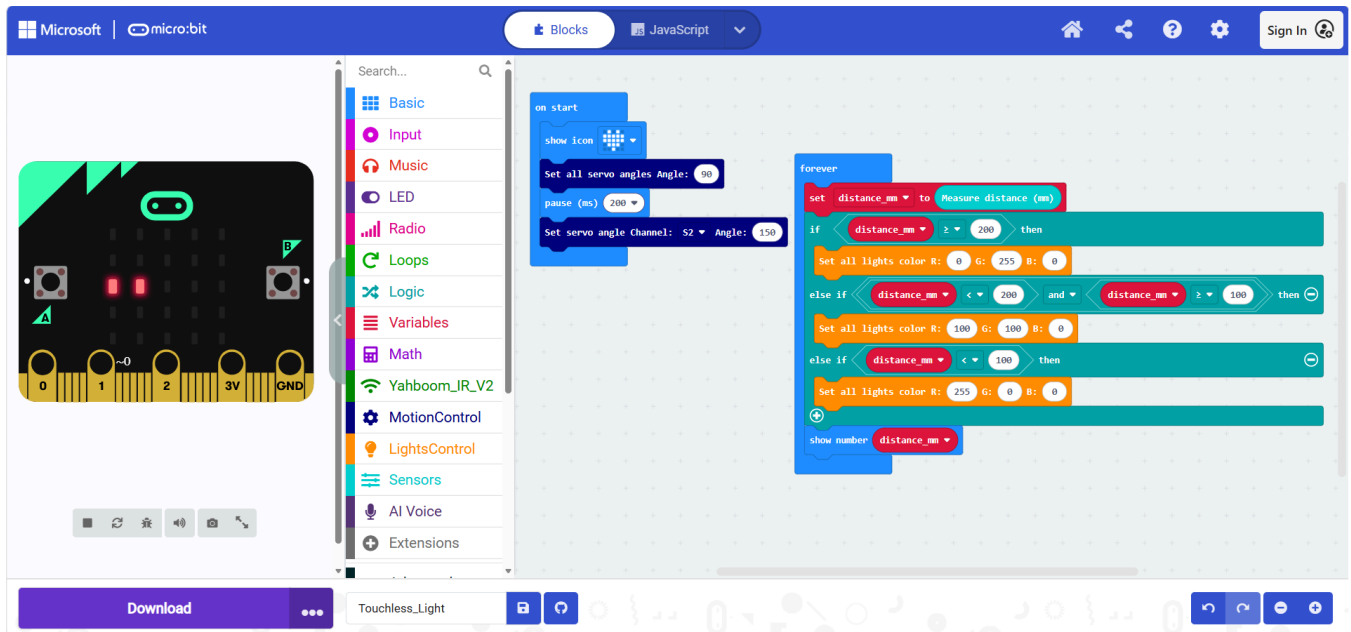
1. Experiment Purpose

In this section, we will use the ultrasonic sensor together with light control to achieve the "touchless light" effect.

2. Implementation Principle



The program is as follows.



The positions of the blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `18.Distance_Control_Light.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

After flashing the program, the ultrasonic sensor detects the distance to the object ahead, and lights of different colors are turned on according to the different distances.

When the object is far from the front of the car, the green light turns on.



When the object is close to the front of the car, the red light turns on.

